

Variational Inference with Implicit Score Priors for Real-World Image Denoising and Beyond

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Abstract Real-world image denoising is practical and crucial in low-level computer vision. Recent score-based generative models have shown effectiveness and promise in zero-shot image restoration. However, applying existing score-based restoration methods requires training explicit score networks on real-world images, and these methods are usually confined to pre-defined generation-oriented procedures, making them inefficient and inflexible. Moreover, existing methods generally assume white Gaussian noise and cannot handle real-world cases. To address these problems, this paper presents an implicit score priors-guided Variational Inference (VI) for flexible and effective real-world denoising. We disclose that the VI framework does not necessitate explicit image priors like traditional treatments. Instead, we propose to guide its variational optimization via implicit score priors, which are obtained through denoising variational Gaussian posteriors' samples via easily available Minimum Mean Squared Error (MMSE) Gaussian denoisers. The proposed optimization utilizes implicit scores as regularization and allows adaptive inference, unleashing explicit score networks and pre-defined samplings. Meanwhile, we exploit a Non-*i.i.d* Gaussian mixture likelihood model and hierarchical Bayesian inference to dynamically model real-world noise during the optimization, thus achieving effective real-world denoising. In addition, a score refinement strategy and a noise-aware prior assignment scheme are developed to stabilize and enhance the optimiza-

tion. Our method provides a novel perspective on variational optimization and is extended to general image restoration, including linear and non-linear restoration tasks. Extensive experiments on diverse tasks and datasets demonstrate the effectiveness and superiority of the proposed method. Code is available at <https://github.com/alwaysuu/ScoreDVI>

Keywords Real-world Denoising · Image Restoration · Variational Inference · Score Priors · MMSE Gaussian denoiser.

1 Introduction

Noise is prevalent in various real-world imaging modalities, including photography imaging [1], fluorescence microscopy imaging [2], and CT imaging [3]. Unlike ideal Additive White Gaussian Noise (AWGN), real-world noise is typically signal-dependent and spatially correlated (structured), which is challenging to remove. For example, photography imaging undergoes intricate image sensor processing pipelines, converting simple raw noise into complex sRGB noise [4]; CT imaging with low radiation transforms noisy projections into images via filtered back projection, thereby complicating CT image noise [3]. As a result, designing effective real-world denoising methods is critical for subsequent image analysis and understanding in practical scenarios.

In the era of deep learning, various DNNs-based supervised [5–8], unsupervised [9,10], and self-supervised methods [11,12,13] have been proposed, which demonstrate promising results on real-world denoising if large-scale training data is available. However, massive data collection is difficult and expensive, and these dataset-based methods generally suffer from model generalization issues [14], limiting their practical applica-

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tion. As an alternative, zero-shot denoising methods that rely on a single noisy image are gaining increasing attention. Nevertheless, existing representative zero-shot denoising methods, like DIP [15], Self2Self [16], ZS-N2N [17], and other works [18–20, 14, 21], show suboptimal performance in denoising real-world images, e.g., natural images, fluorescence microscopy images and CT images, necessitating more suitable approaches.

Recent score (and diffusion)-based generative models have exhibited remarkable distribution modeling capacity over various signals [22–26]. Building on massive high-quality images and large DNNs, score-based models generate realistic and high-fidelity images [22, 24], demonstrating that characterizing a distribution’s score, i.e., the gradient of its logarithm, is a promising and superior way to model natural images’ distribution (also the image prior). Furthermore, incorporating degraded measurements into the sampling pipeline of score-based models enables conditional image generation and motivates various *zero-shot* restoration methods [27–34]. Despite the great progress, applying the above generation-oriented methods to restoration requires collecting massive high-quality images from target modalities and then training large score networks [35, 22], which is usually *expensive*. Even with well-trained score networks, these methods must follow the pre-designed generative procedure of score-based models, e.g., DDPM [23] and DDIM [36] sampling used in [27–29], to achieve good image restoration, meaning that learned score priors are employed in a *restricted* way. In addition, many score-based zero-shot restorations consider AWGN and *perform poorly* in real-world cases. Some subsequent variants [37, 38] focus on blind degradations but primarily examine unknown forward operators (e.g., blurring, downsampling) rather than noise, resulting in inferior real-world denoising.

To address these limitations, we propose implicit score priors-driven Variational Inference (VI), namely ScoreDVI+, for flexible real-world zero-shot denoising and other restoration tasks. First, to avoid training explicit score networks, we leverage Tweedie’s formula [39], an amazing equation relating the score of a univariate distribution to the MMSE estimate of its AWGN-corrupted version, and extend the formula to the multivariate case for high-dimensional images. By inspecting the Evidence Lower Bound (ELBO) objective in VI, we uncover that the variational optimization can be carried out by utilizing the image prior’s score, which is implicitly obtained from denoising the Gaussian posterior’s samples with off-the-shelf MMSE Gaussian denoisers. Gradients from the derived score update variational parameters, which then allow dynamical score extraction. This iterative procedure applies *cheap* scores from eas-

ily available Gaussian denoisers and achieves *adaptive* optimization, without relying on explicit priors or pre-defined sampling in score-based methods.

To handle real-world complex noises, we represent the observation likelihood via a Non-*i.i.d.* Gaussian Mixture Model (GMM), which is capable of modeling intricate and spatially varying signals [40] and does not introduce significant computational burden. Meanwhile, a variational precision posterior is constructed and inferred during optimization to handle unclear noise strengths, achieving a *dynamical update* of likelihood and hence effective real-world denoising. The employed GMM also allows fusing multiple score priors and enhances denoising capability. Additionally, we propose a noise-aware prior assignment scheme that assigns diverse portions of score priors to different real-world images, thereby further enhancing the performance of individual instances.

Beyond denoising, we robustify the optimization process of our methods and adapt it to other image restoration tasks. Recognizing that off-the-shelf Gaussian denoisers may introduce unstable scores during variational optimization, we devise a simple yet effective score refinement strategy, which utilizes the image-level (i.e., average) variance of the variational image posterior rather than dense posterior variance to extract scores, resulting in stable gradient updates. To handle linear and non-linear image restoration, we incorporate linear system matrices and non-linear forward operators into the EBLO and derive the corresponding loss functions for diverse restoration tasks under the proposed score-driven VI framework, extending the application scope of our approach.

Finally, we comprehensively evaluate our method on diverse tasks, including denoising, deblurring, inpainting, HDR, and JPEG restoration. For real-world denoising, several representative imaging modalities are considered, i.e., photography images, fluorescence microscopy images, and CT images. The proposed method outperforms other zero-shot methods in various benchmark datasets and achieves competitive results against dataset-based methods. Regarding deblurring, our method surpasses popular DDRM [27] and DiffPIR [30] when removing isotropic and anisotropic Gaussian blurs, and outperforms DPS [29] and RED-diff [41] on motion deblurring. The proposed method is also effective for non-linear HDR and JPEG restoration.

To summarize, our contributions are as follows:

1. We propose a novel variational optimization framework, driven by implicit scores derived from off-the-shelf MMSE Gaussian denoisers. The proposed approach provides a general paradigm for VI.

Table 1: The survey table of related works

Related works	Category 1	Category 2	References
Un/self-supervised denoising methods	Dataset-based	Regularization-based	[42, 43]
		Resampling-based	[44–46]
		Blind-spot-based	[18, 47–50, 1, 13, 12]
		Unpaired/Disentanglement	[10, 9, 11]
	Zero-shot	External priors	[15, 14, 51, 52]
Priors-based Image Restoration	Classical priors	Self-supervision	[16, 17, 20, 53, 21, 54, 55]
		–	[56–59]
	Score-based models	General methods	[27–30, 33, 31, 41]
		Imaging specific	[32, 34]
VI	Traditional VI	Real-world/complex degradation	[60–62, 37, 38]
		–	[63–65]
	Deep VI with paired data	–	[66–68]

2. We integrate a Non-*i.i.d.* GMM likelihood and hierarchical Bayesian inference into the optimization to better accommodate real-world noise. In addition, a noise-aware prior assignment scheme is developed.
3. We introduce a robust score refinement strategy to enhance the stability of variational optimization.
4. We extend the proposed framework to a variety of linear and non-linear restoration tasks, demonstrating its versatility and general applicability.

A preliminary version of this work, namely ScoreDVI, was presented in [69] as a conference paper. Compared with ScoreDVI, this paper introduces three additional key contributions: 1) A score refinement strategy is proposed to stabilize score estimation during variational optimization; 2) ScoreDVI is extended to diverse linear and non-linear restoration tasks, including deblurring, inpainting, HDR, and JPEG restoration, thereby improving its practicality; 3) Additional real-world low-dose CT denoising experiments are conducted to validate the generalization capability of the proposed method.

2 Related Works

This section reviews recent methods related to our paper, including real-world denoising, score-based image restoration, and deep-learning-based VI. The summary of these related works is presented in Table 1.

2.1 Real-world Image Denoising

Dataset-based un/self-supervised methods. As supervised learning methods [70, 6–8, 71, 72] and Noise2Noise [73] suffer from paired-data collection difficulties in practice, many self-supervised methods without using paired data have been proposed. Regularization-based methods like Noise2Score [42] and Stein-based work [43] for several specific noise

types like AWGN and Poisson noise. Blind-spot (BS) based methods, e.g., Noise2self [18], Neighbor2neighbor [47], B2U [48] and BS networks [49, 50], are effective in removing spatially independent noise. Resampling-based methods, e.g., Noiser2noise [44], R2R [45], and IDR [46], can address structured noises when providing accurate noise distributions. However, the above methods cannot deal with real-world photography or CT images, as real-world noise is signal-dependent and spatially correlated, which is generally structured and hard to estimate [1].

To search for supervision from the real-world image itself, Pixel-shuffle Downsampling (PD) [19] and neighboring mask convolution [12] are proposed to extend BS to real-world noise. PD operation breaks down the spatial correlation of structured noise, allowing training BS networks on sub-sampled images. Representative works along this line are AP-BSN [1] and PUCA [13]. Instead, the latter excludes neighboring pixels that correlate noise with the center pixel via masked convolution [12]. In addition, several unpaired methods [10, 50] and disentanglement-based methods [9, 11] are proposed to tackle general denoising tasks. Nevertheless, these methods merely perform well when given massive noisy or unpaired data, and their performance degrades in data scarcity cases, as shown in our experiments.

Zero-shot denoising. Deep learning-based zero-shot denoising methods rely on either external priors or self-supervision. Regarding the former, DIP [15] achieved zero-shot denoising by utilizing the implicit regularization of CNN architectures. NN+denoiser [14] used a conditional VAE to model the likelihood of real-world noise and combined plug-and-play denoisers [51, 52] to denoise. For the latter, Self2Self [16] built training pairs from a single noisy image via Bernoulli sampling to train denoisers, while ZS-N2N [17] used PD. StructN2V [20] estimated the noise correlation from a single image and developed a structured mask to apply N2V [53]. However, these methods generally show suboptimal performance when removing complex and severe

real-world noise. Recently, some variants have demonstrated improved results on real-world denoising [54, 55]. In this paper, we incorporate implicit score priors from MMSE Gaussian denoisers into VI and meanwhile introduce Non-*i.i.d.* GMM likelihood with hierarchical Bayesian inference, significantly improving the denoising quality of real-world noisy images.

2.2 Score-based Image Restoration

Classical denoising methods [56–59] utilize local or patch statistics of natural images to denoise. Recently, global image statistics can be learned via score (or diffusion)-based deep generative models [22–24]. These methods generate realistic images after training on large-scale, high-quality datasets. Combining Bayesian inference and the degradation process, score-based generative models are ready for zero-shot image restoration, and various related methods have been proposed, in which popular methods include DDRM [27], DDNM [28], DPS [29], DiffPIR [30], and so on [32, 33, 31, 34].

However, the above score-based restoration methods generally consider degraded images with AWGN and cannot effectively remove real-world noise. Recently, several works have particularly focused on real-world or complex degradations, including [60–62, 37, 38], among which DiffusionVI [61] is the best denoising method that conducts likelihood estimation during the generation process. Despite the progress, score-based restoration requires explicit score networks that may not be easy to train for specific image modalities, e.g., CT images. Furthermore, real-world denoising under such a generation-based framework must strictly adhere to the predefined generation process, which is inflexible and limited. Instead, this paper proposes an adaptive optimization driven by score priors extracted from blind Gaussian denoisers. The optimization of our method is adaptive and unleashes generation paradigms.

2.3 Deep learning-based Variational Inference

Traditional VI for image restoration [63–65] employ analytical priors and iterative optimization to obtain the approximate posterior, but they often suffer from poor performance. Recently, deep learning-based VI methods for real-world image denoising, such as VDN [66], VNIR [67], and VDIR [68] construct variational posteriors via DNNs and utilize ground-truth images as strong priors. However, these methods require paired training data. Instead, our method is unsupervised and can deal with real-world noise by employing implicit score priors from blind Gaussian denoisers.

3 Preliminary

3.1 Variational Inference for Image Denoising

In Bayesian statistics, image denoising is viewed as finding the posterior distribution $p(\mathbf{x}|\mathbf{y})$ and its estimators, e.g., posterior mean or MAP solution, where $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{N \times 1}$ are vectorized clean image and noisy observation, respectively. In practice, the posterior distribution $p(\mathbf{x}|\mathbf{y})$ is non-trivial and has no closed-form solution. The idea of VI is to construct a tractable variational distribution $q(\mathbf{x})$ and minimize its ‘distance’ to the true $p(\mathbf{x}|\mathbf{y})$. KL divergence is usually employed as the distance metric:

$$\begin{aligned} \text{KL}(q(\mathbf{x})||p(\mathbf{x}|\mathbf{y})) &= \int \log \frac{q(\mathbf{x})}{p(\mathbf{x}|\mathbf{y})} q(\mathbf{x}) d\mathbf{x} \\ &= \log p(\mathbf{y}) - \underbrace{(\mathbb{E}_{q(\mathbf{x})}(\log p(\mathbf{y}|\mathbf{x})) - \text{KL}(q(\mathbf{x})||p(\mathbf{x})))}_{\text{ELBO}} \end{aligned} \quad (1)$$

As $\log p(\mathbf{y})$ is constant, ELBO in Eq. (1) is used as the loss function of VI. Modeling the image prior $p(\mathbf{x})$ and likelihood $p(\mathbf{y}|\mathbf{x})$, and then inferring variational posterior $q(\mathbf{x})$ is the standard procedure of VI.

3.2 Tweedie’s formula

Tweedie’s formula [39] states that, given univariate $x \sim p(x)$ and $p(\tilde{x}|x) \sim \mathcal{N}(x, \sigma^2)$, the score of $p(\tilde{x})$ is given by the following equation:

$$\nabla_{\tilde{x}} \log p(\tilde{x}) = \frac{\mathbb{E}(x|\tilde{x}) - \tilde{x}}{\sigma^2} \quad (2)$$

where $\mathbb{E}(x|\tilde{x})$ is the MMSE estimate of x given the observation $\tilde{x}$ corrupted by AWGN.

4 Method

This section presents our implicit score-driven VI for real-world denoising. In section 4.1, we explain the methodology of applying score priors for variational optimization. The Non-*i.i.d.* Gaussian mixture likelihood model and hyperpriors are introduced in section 4.2. Section 4.3 shows variational posteriors and the complete optimization objective. The noise-aware prior assignment is given in section 4.4.

4.1 Implicit Score Priors for Variational Inference

4.1.1 Extended Tweedie’s formula

Here, we first extend Eq. (2) to the multi-variate case:

Corollary 1 Suppose $p(\tilde{\mathbf{x}}|\mathbf{x}) \sim \mathcal{N}(\mathbf{x}, \Sigma)$ and $\mathbf{x} \sim p(\mathbf{x})$. Then the score of $p(\tilde{\mathbf{x}})$ is

$$\nabla_{\tilde{\mathbf{x}}} \log p(\tilde{\mathbf{x}}) = \Sigma^{-1} (\mathbb{E}(\mathbf{x}|\tilde{\mathbf{x}}) - \tilde{\mathbf{x}}) \quad (3)$$

where $\mathbb{E}(\mathbf{x}|\tilde{\mathbf{x}})$ is the MMSE estimate of $\mathbf{x}$ given $\tilde{\mathbf{x}}$ with multi-variate Gaussian noise.

Proof First, the marginal distribution $p(\tilde{\mathbf{x}}) = \mathbb{E}_{p(\mathbf{x})}[p(\tilde{\mathbf{x}}|\mathbf{x})] = \int \mathcal{N}(\mathbf{x}, \Sigma) p(\mathbf{x}) d\mathbf{x}$. The gradient of $p(\tilde{\mathbf{x}})$ is computed as follows:

$$\begin{aligned} \nabla_{\tilde{\mathbf{x}}} p(\tilde{\mathbf{x}}) &= \nabla_{\tilde{\mathbf{x}}} \mathbb{E}_{p(\mathbf{x})} \left[Z \exp \left\{ -\frac{1}{2} (\tilde{\mathbf{x}} - \mathbf{x})^T \Sigma^{-1} (\tilde{\mathbf{x}} - \mathbf{x}) \right\} \right] \\ &= \int \left(Z \exp \left\{ -\frac{1}{2} (\tilde{\mathbf{x}} - \mathbf{x})^T \Sigma^{-1} (\tilde{\mathbf{x}} - \mathbf{x}) \right\} \times \right. \\ &\quad \left. \nabla_{\tilde{\mathbf{x}}} \left(-\frac{1}{2} (\tilde{\mathbf{x}} - \mathbf{x})^T \Sigma^{-1} (\tilde{\mathbf{x}} - \mathbf{x}) \right) \right) p(\mathbf{x}) d\mathbf{x} \\ &= \int \Sigma^{-1} (\mathbf{x} - \tilde{\mathbf{x}}) p(\tilde{\mathbf{x}}|\mathbf{x}) p(\mathbf{x}) d\mathbf{x} \\ &= \Sigma^{-1} \int (\mathbf{x} - \tilde{\mathbf{x}}) p(\tilde{\mathbf{x}}, \mathbf{x}) d\mathbf{x} \\ &= \Sigma^{-1} \int (\mathbf{x} - \tilde{\mathbf{x}}) p(\mathbf{x}|\tilde{\mathbf{x}}) p(\tilde{\mathbf{x}}) d\mathbf{x} \end{aligned} \quad (4)$$

where $Z = \frac{1}{(2\pi)^{0.5N} |\Sigma|^{0.5}}$ is the normalization coefficient.

Multiplying both sides of Eq. (4) by $\frac{1}{p(\tilde{\mathbf{x}})}$ gives

$$\begin{aligned} \frac{\nabla_{\tilde{\mathbf{x}}} p(\tilde{\mathbf{x}})}{p(\tilde{\mathbf{x}})} &= \nabla_{\tilde{\mathbf{x}}} \log p(\tilde{\mathbf{x}}) = \Sigma^{-1} \int (\mathbf{x} - \tilde{\mathbf{x}}) p(\mathbf{x}|\tilde{\mathbf{x}}) d\mathbf{x} \\ &= \Sigma^{-1} (\mathbb{E}(\mathbf{x}|\tilde{\mathbf{x}}) - \tilde{\mathbf{x}}) \end{aligned} \quad (5)$$

where $\mathbb{E}(\mathbf{x}|\tilde{\mathbf{x}})$ is the conditional mean for noisy input $\tilde{\mathbf{x}}$ corrupted by Gaussian noise. The proof is completed.

If we further consider Σ has diagonal form, i.e., $\Sigma = \text{diag}(\sigma^2)$, Eq. (3) becomes to:

$$\nabla_{\tilde{\mathbf{x}}} \log p(\tilde{\mathbf{x}}) = (\mathbb{E}(\mathbf{x}|\tilde{\mathbf{x}}) - \tilde{\mathbf{x}}) \odot \frac{1}{\sigma^2} \quad (6)$$

where $\odot$ denotes element-wise multiplication.

Note that the MMSE estimate $\mathbb{E}(\mathbf{x}|\tilde{\mathbf{x}})$ is the conditional expectation of the posterior $p(\mathbf{x}|\tilde{\mathbf{x}})$, which meanwhile is the target of supervised learning-based deep denoisers under the MSE loss [73, 74]. That is, $\mathbb{E}(\mathbf{x}|\tilde{\mathbf{x}})$ in Eq. (6) can be replaced with the output of an ideal MMSE Gaussian denoiser. Moreover, we can use a blind Gaussian denoiser $\mathcal{G}_\theta(\cdot)$ trained over AWGN with a wide range of noise variance to approximate the ideal one due to the local property of CNNs. Therefore, score priors are directly available from well-trained blind Gaussian denoisers, and there is no need to explicitly train score networks like done in score-based generative models [75]. That is,

$$\nabla_{\tilde{\mathbf{x}}} \log p(\tilde{\mathbf{x}}) = (\mathcal{G}_\theta(\tilde{\mathbf{x}}) - \tilde{\mathbf{x}}) \odot \frac{1}{\sigma^2} \quad (7)$$

where practical $\mathcal{G}_\theta(\tilde{\mathbf{x}})$ substitutes for ideal $\mathbb{E}(\mathbf{x}|\tilde{\mathbf{x}})$ in Eq. (6).

4.1.2 Score facilitates variational optimization

Within the ELBO objective in Eq. (1), the image prior $p(\mathbf{x})$ is only involved in its KL term, i.e., $\text{KL}(q(\mathbf{x})||p(\mathbf{x}))$, which can be approximated as:

$$\begin{aligned} \text{KL}(q(\mathbf{x})||p(\mathbf{x})) &= \mathbb{E}_{q(\mathbf{x})} \log q(\mathbf{x}) - \mathbb{E}_{q(\mathbf{x})} \log p(\mathbf{x}) \\ &\approx \mathbb{E}_{q(\mathbf{x})} \log q(\mathbf{x}) - \frac{1}{M} \sum_{m=1}^M \log p(\mathbf{x}^m) \end{aligned} \quad (8)$$

with $\mathbf{x}^m \sim q(\mathbf{x})$

where Monte Carlo (MC) integration with M samples is used to approximate $\mathbb{E}_{q(\mathbf{x})} \log p(\mathbf{x})$.

We can find that if we take the derivative of $\log p(\mathbf{x}^m)$ w.r.t. $\mathbf{x}^m$, the score term $\nabla_{\mathbf{x}^m} \log p(\mathbf{x}^m)$ appears. When we further consider $q(\mathbf{x})$ follows diagonal Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2))$, a common practice in VI, then the derivatives of $\mathbb{E}_{q(\mathbf{x})} \log p(\mathbf{x})$ w.r.t. variational posterior parameters $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}^2$ can be obtained. Particularly, Based on the MC integration in Eq. (8) and chain rule, $\log p(\mathbf{x}^m)$'s gradient w.r.t. $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ is computed as follows:

$$\begin{aligned} \nabla_{\boldsymbol{\mu}} \mathbb{E}_{q(\mathbf{x})} \log p(\mathbf{x}) &\approx \nabla_{\boldsymbol{\mu}} \frac{1}{M} \sum_{m=1}^M \log p(\mathbf{x}^m) \\ &= \frac{1}{M} \sum_{m=1}^M \nabla_{\mathbf{x}^m} \log p(\mathbf{x}^m) \odot \nabla_{\boldsymbol{\mu}} \mathbf{x}^m \\ &= \frac{1}{M} \sum_{m=1}^M \nabla_{\mathbf{x}^m} \log p(\mathbf{x}^m) \end{aligned} \quad (9)$$

$$\begin{aligned} \nabla_{\boldsymbol{\sigma}} \mathbb{E}_{q(\mathbf{x})} \log p(\mathbf{x}) &\approx \nabla_{\boldsymbol{\sigma}} \frac{1}{M} \sum_{m=1}^M \log p(\mathbf{x}^m) \\ &= \frac{1}{M} \sum_{m=1}^M \nabla_{\mathbf{x}^m} \log p(\mathbf{x}^m) \odot \nabla_{\boldsymbol{\sigma}^2} \mathbf{x}^m \\ &= \frac{1}{M} \sum_{m=1}^M \nabla_{\mathbf{x}^m} \log p(\mathbf{x}^m) \odot \boldsymbol{\epsilon} \end{aligned} \quad (10)$$

where we make use the fact that $\mathbf{x}^m = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I)$.

After replacing scores in above two equations, i.e., $\nabla_{\mathbf{x}^m} \log p(\mathbf{x}^m)$, with $(\mathcal{G}_\theta(\mathbf{x}^m) - \mathbf{x}^m) \odot \frac{1}{\sigma^2}$, i.e., the practical implementation in Eq. (7), we can obtain the following derivatives:

$$\begin{aligned} \nabla_{\boldsymbol{\mu}} \mathbb{E}_{q(\mathbf{x})} \log p(\mathbf{x}) &\approx \frac{1}{M} \sum_{m=1}^M (\mathcal{G}_\theta(\mathbf{x}^m) - \mathbf{x}^m) \odot \frac{1}{\sigma^2} \\ \nabla_{\boldsymbol{\sigma}} \mathbb{E}_{q(\mathbf{x})} \log p(\mathbf{x}) &\approx \frac{1}{M} \sum_{m=1}^M (\mathcal{G}_\theta(\mathbf{x}^m) - \mathbf{x}^m) \odot \frac{\boldsymbol{\epsilon}}{\sigma^2} \end{aligned} \quad (11)$$

$$\mathbf{x}^m \sim q(\mathbf{x}) \Rightarrow \mathbf{x}^m = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}, \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I)$$

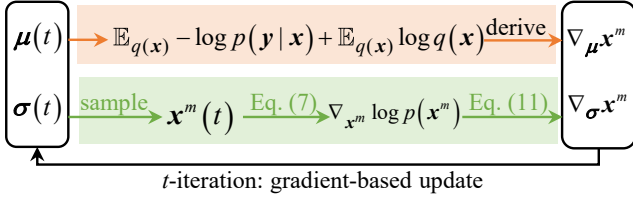


Fig. 1: Illustration of variational optimization based on the implicit score from blind Gaussian denoisers.

Eq. (11) reveals that there is no need to explicitly design the prior $p(\mathbf{x})$ and then compute $\mathbb{E}_{q(\mathbf{x})} \log p(\mathbf{x})$ analytically. Instead, we can utilize scores from off-the-shelf blind Gaussian denoisers to derive gradients of variational parameters $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$, directly injecting score priors into the optimization of $\mathbb{E}_{q(\mathbf{x})} \log p(\mathbf{x})$. Regarding $\mathbb{E}_{q(\mathbf{x})} \log q(\mathbf{x})$ in the ELBO, they can be analytically derivative to $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ by design. We show a diagram of variational parameter update with implicit score priors in Fig. 1.

We note that extracting scores from Gaussian denoisers and updating $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are adaptive during the ELBO optimization, and we do not need to pre-define the noise variance and design the sampling strategy as done in score-based restoration methods [22, 31, 27]. That is, the score in the $(t-1)$ -th iteration assists in the update of $\boldsymbol{\mu}(t)$ and $\boldsymbol{\sigma}(t)$, which subsequently generates new $\mathbf{x}^m(t)$ for extracting next score in the t -th iteration.

4.1.3 Score refinement for robust training

In practice, using a blind Gaussian denoiser $\mathcal{G}_\theta(\mathbf{x}^m)$ to approximate the ideal MMSE estimate inevitably introduces errors. For example, off-the-shelf $\mathcal{G}_\theta(\cdot)$ may overly or insufficiently denoise $\mathbf{x}^m$ in different spatial locations. As a result, the denoising residual $\mathcal{G}(\mathbf{x}^m) - \mathbf{x}^m$ may misalign with underlying variational posterior variance $\boldsymbol{\sigma}^2$. As a comparison, an ideal Gaussian denoiser induces a residual that exactly follows $\mathcal{N}(0, \text{diag}(\boldsymbol{\sigma}^2))$. During the variational optimization, the division of these mismatched residuals and variational variances will lead to a sudden amplification of scores, misleading the gradient update. Such a phenomenon is visualized in Fig. 2 (and also verified in Table 11 of Section 6), where the posterior variance $\boldsymbol{\sigma}^2$ and the extracted score show a fluctuation during the optimization process (see the blue plots around 400-th iteration).

To address this issue, we propose a simple yet effective score refinement strategy. Instead of using the dense variance map $\boldsymbol{\sigma}^2$ to derive scores as indicated in Eq. (7), we utilize its spatial average $\bar{\boldsymbol{\sigma}^2}$ to prevent situations in which unreasonably small or large variance

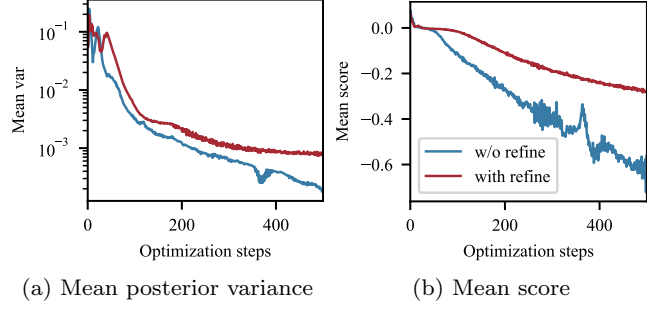


Fig. 2: Variational optimization with (using Eq. (12)) and without score refinement (using Eq. (7)).

elements in $\boldsymbol{\sigma}^2$ cause unstable score and hence mislead optimization. That is, the refined score is extracted as:

$$\nabla_{\mathbf{x}^m} \log p(\mathbf{x}^m) \approx (\mathcal{G}_\theta(\mathbf{x}^m) - \mathbf{x}^m) \odot \frac{1}{\bar{\boldsymbol{\sigma}^2}} \quad (12)$$

As shown by the red plots in Fig. 2, the average score under this refinement strategy decreases steadily, and variational optimization becomes more stable. The effectiveness of this refinement strategy is ablated in Table 11 of Section 6.

4.2 Non-*i.i.d* Gaussian Mixture Likelihood Model

To utilize the VI framework and Eq. (11) for real-world denoising, it's crucial to model the likelihood function $p(\mathbf{y}|\mathbf{x})$. It is well known that real-world image noise is signal-dependent and spatially correlated, which differs greatly from AWGN and is challenging to model precisely. DiffusionVI shows that a heterogeneous and structured Gaussian model can roughly characterize real-world noise, but suffers from difficulties in parameter estimation and matrix calculation [76]. Therefore, we resort to approximating such complex noise with a tractable model that captures its main features and facilitates ELBO optimization. Here, we employ an independent but not identically distributed (i.e., Non-*i.i.d.*) Gaussian mixture model (GMM) to model the likelihood function:

$$p(\mathbf{y}|X, \Phi, \Omega) = \prod_{i=1}^N \sum_{k=1}^K \omega_{ik} \mathcal{N}(x_{ik}, \phi_{ik}^{-1}) \quad (13)$$

where $\sum_{k=1}^K \omega_{ik} = 1$; $X = [\mathbf{x}_1, \dots, \mathbf{x}_K]$, $\Phi = [\phi_1, \dots, \phi_K]$, $\Omega = [\omega_1, \dots, \omega_K] \in \mathbb{R}^{N \times K}$ are matrices with K columns, respectively; x_{ik} in X , ϕ_{ik} in Φ , and ω_{ik} in Ω denote the mean, precision, and mixing parameter of the k -th Gaussian component, respectively, for modeling i -th pixel y_i of $\mathbf{y}$; K is the total number of mixtures.

In Eq. (13), each noisy pixel y_i is represented by an individual GMM, and different y_i are separately modeled, to capture the spatially variant characteristic of real-world noise. Besides, such consideration allows the ensemble of different priors and variational image posteriors, which will be introduced in the following section. To facilitate optimization, we refer to [40] and reparameterize the likelihood function in Eq. (13) as:

$$\begin{aligned} p(\mathbf{y}|X, Z, \Phi) &= \prod_{i=1}^N \prod_{k=1}^K \mathcal{N}(x_{ik}, \phi_{ik}^{-1})^{z_{ik}} \\ p(Z|\Omega) &= \prod_{i=1}^N \prod_{k=1}^K \omega_{ik}^{z_{ik}}, \sum_{k=1}^K z_{ik} = 1 \end{aligned} \quad (14)$$

where $Z = [z_1, \dots, z_K] \in \mathbb{R}^{N \times K}$ is an auxiliary matrix variable and each row of Z follows a categorical distribution with one item one and the others zero. It is easy to verify that $p(\mathbf{y}|X, \Phi, \Omega) = \sum_Z p(\mathbf{y}|X, \Phi, Z)p(Z|\Omega)$.

Different from many score-based methods that carry out posterior inference based on accurate likelihood functions, which is impossible for real-world noise, we view Φ and Ω in Eq. (14) as random variables and infer them by introducing hyperpriors based on the hierarchical Bayesian framework, allowing dynamical estimation of likelihood. To simplify the ELBO objective, we choose the following conjugate priors for Φ and Ω , respectively:

$$\begin{aligned} p(\Phi|Z) &= \prod_{i=1}^N \prod_{k=1}^K \text{Gamma}(\phi_{ik}; \alpha, \beta)^{z_{ik}} \\ p(\Omega) &= \prod_{i=1}^N Z_{\text{Dir}}(\mathbf{d}) \prod_{k=1}^K \omega_{ik}^{d_k-1}, \mathbf{d} \in \mathbb{R}^K \end{aligned} \quad (15)$$

where Gamma denotes the Gamma distribution with scalar parameters α and β . Each row of Ω follows the Dirichlet distribution with parameter $\mathbf{d}$ and normalization coefficient $Z_{\text{Dir}}(\cdot)$.

4.3 Variational Optimization for Real-world Denoising

4.3.1 Constructing variational posteriors

To align with Eq. (14), we consider the factorized image prior conditioned on Z for real-world denoising:

$$p(X|Z) = \prod_{i=1}^N \prod_{k=1}^K p(x_{ik})^{z_{ik}} \quad (16)$$

which enables the prior $p(X)$ to be modeled by fusing different priors pixel-wise.

Given Eqs. (14), (15) and (16), the posterior we aim to infer for real-world denoising is $p(X, \Phi, Z, \Omega|\mathbf{y})$,

which is non-trivial. As a result, we design a tractable variational posterior $q(X, \Phi, Z, \Omega)$ to approximate the true $p(X, \Phi, Z, \Omega|\mathbf{y})$. Specifically, we factorize the variational posterior $q(X, \Phi, Z, \Omega)$ as $q(X|Z)q(\Phi|Z)q(Z)q(\Omega)$ with each item as:

$$\begin{aligned} q(X|Z) &= \prod_{i=1}^N \prod_{k=1}^K \mathcal{N}(\mu_{ik}, \sigma_{ik}^2)^{z_{ik}} \\ q(\Phi|Z) &= \prod_{i=1}^N \prod_{k=1}^K \text{Gamma}(\phi_{ik}; \hat{\alpha}_{ik}, \hat{\beta}_{ik})^{z_{ik}} \\ q(Z) &= \prod_{i=1}^N \prod_{k=1}^K \pi_{ik}^{z_{ik}}, q(\Omega) = \prod_{i=1}^N Z_{\text{Dir}}(\hat{\mathbf{d}}_i) \prod_{k=1}^K \omega_{ik}^{\hat{d}_{ik}-1} \end{aligned} \quad (17)$$

All variational posteriors' parameters in Eq. (17) are denoted as $\Theta = \{\mu_{ik}, \sigma_{ik}, \hat{\alpha}_{ik}, \hat{\beta}_{ik}, \pi_{ik}, \hat{d}_{ik} | i \in \{1, \dots, N\}, k \in \{1, \dots, K\}\}$. Our objective is to optimize Θ under the guidance of implicit score priors and the GMM-based noise model.

4.3.2 Deriving loss function

Analogous to Eq. (1), we can derive the complete loss function for optimizing Θ under the real-world denoising case:

$$\mathcal{L}_{\text{denoise}} = -\text{ELBO} \quad (18a)$$

$$= -\mathbb{E}_{q(X, \Phi, Z)}(\log p(\mathbf{y}|X, \Phi, Z)) \rightarrow \mathcal{L}_1 \quad (18b)$$

$$+ \mathbb{E}_{q(Z)}(\text{KL}(q(X|Z)||p(X|Z))) \rightarrow \mathcal{L}_2 \quad (18c)$$

$$+ \mathbb{E}_{q(Z)}(\text{KL}(q(\Phi|Z)||p(\Phi|Z))) \rightarrow \mathcal{L}_3 \quad (18d)$$

$$+ \mathbb{E}_{q(\Omega)}(\text{KL}(q(Z)||p(Z|\Omega))) \rightarrow \mathcal{L}_4 \quad (18e)$$

$$+ \text{KL}(q(\Omega)||p(\Omega)) \rightarrow \mathcal{L}_5 \quad (18f)$$

with each component as follows:

$$\mathcal{L}_1 = -\sum_{i=1}^N \sum_{k=1}^K \pi_{ik} \left(-\frac{((\mu_{ik} - y_i)^2 + \sigma_{ik}^2) \hat{\alpha}_{ik}}{2\hat{\beta}_{ik}} + \frac{\psi(\hat{\alpha}_{ik}) - \log \hat{\beta}_{ik}}{2} \right) \quad (19)$$

$$\mathcal{L}_2 = \sum_{i=1}^N \sum_{k=1}^K \pi_{ik} \left(-\frac{1}{2} \log \sigma_{ik}^2 - \mathbb{E}_{q(x_{ik})} \log p(x_{ik}) \right) \quad (20)$$

$$\begin{aligned} \mathcal{L}_3 &= \sum_{i=1}^N \sum_{k=1}^K \pi_{ik} \left(\log \frac{\Gamma(\alpha)}{\Gamma(\hat{\alpha}_{ik})} + (\hat{\alpha}_{ik} - \alpha) \psi(\hat{\alpha}_{ik}) \right. \\ &\quad \left. + \alpha \log \frac{\hat{\beta}_{ik}}{\beta} + (\beta - \hat{\beta}_{ik}) \frac{\hat{\alpha}_{ik}}{\hat{\beta}_{ik}} \right) \end{aligned} \quad (21)$$

$$\mathcal{L}_4 = \sum_{i=1}^N \sum_{k=1}^K \pi_{ik} \left(\log \pi_{ik} - \psi(\hat{d}_{ik}) + \psi\left(\sum_{k=1}^K \hat{d}_{ik}\right) \right) \quad (22)$$

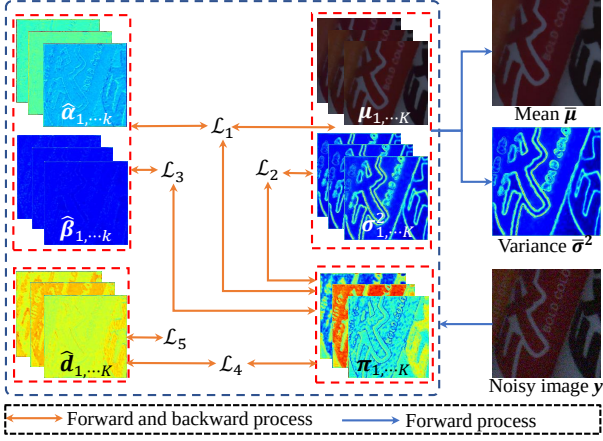


Fig. 3: Relationship between parameters Θ and loss functions $\mathcal{L}$ in Eq. (18) under real-world denoising.

$$\begin{aligned} \mathcal{L}_5 = & \sum_{i=1}^N \sum_{k=1}^K (\hat{d}_{ik} - d_k) (\psi(\hat{d}_{ik}) - \psi(\sum_{k=1}^K \hat{d}_{ik})) \\ & + \sum_{i=1}^N \log \frac{Z_{\text{Dir}}(\hat{\mathbf{d}}_i)}{Z_{\text{Dir}}(\mathbf{d})} \end{aligned} \quad (23)$$

where $\Gamma(\cdot)$ and $\psi(\cdot)$ denote gamma function and digamma function, respectively. The detailed derivation is given in Supplementary Material (SM) S1.

Note that $\mathcal{L}_1, \mathcal{L}_3, \mathcal{L}_4, \mathcal{L}_5$ and $-\frac{1}{2}\pi_{ik} \log \sigma_{ik}^2$ in $\mathcal{L}_2$ are all directly computable and analytically derivable. Regarding $\mathbb{E}_{q(x_{ik})} \log p(x_{ik})$ in $\mathcal{L}_2$, we highlight that it can be *implicitly optimized via scores*, as introduced in Section 4.1. Due to the factorized $p(X|Z)$ and $q(X|Z)$, we have

$$\begin{aligned} \nabla_{\mu_{ik}} \mathbb{E}_{q(x_{ik})} \log p(x_{ik}) &= (\nabla_{\mu_k} \mathbb{E}_{q(\mathbf{x}_k)} \log p(\mathbf{x}_k))_i \\ \nabla_{\sigma_{ik}} \mathbb{E}_{q(x_{ik})} \log p(x_{ik}) &= (\nabla_{\sigma_k} \mathbb{E}_{q(\mathbf{x}_k)} \log p(\mathbf{x}_k))_i \end{aligned} \quad (24)$$

which is computed by combining Eqs. (11) and (12):

$$\begin{aligned} \nabla_{\mu_k} \mathbb{E}_{q(\mathbf{x}_k)} \log p(\mathbf{x}_k) &\approx \frac{1}{M} \sum_{m=1}^M (\mathcal{G}_{\theta}^k(\mathbf{x}_k^m) - \mathbf{x}_k^m) \odot \frac{1}{\sigma_k^2} \\ \nabla_{\sigma_k} \mathbb{E}_{q(\mathbf{x}_k)} \log p(\mathbf{x}_k) &\approx \frac{1}{M} \sum_{m=1}^M (\mathcal{G}_{\theta}^k(\mathbf{x}_k^m) - \mathbf{x}_k^m) \odot \frac{1}{\sigma_k^2} \odot \epsilon \end{aligned} \quad (25)$$

where $\mathcal{G}_{\theta}^k(\cdot)$ is the k -th employed blind Gaussian denoiser, from which we recognize that the GMM-based likelihood allows us to incorporate multiple scores into the optimization, acting as the priors' ensemble.

The optimization pipeline of the proposed method is presented in Alg. 1, and the role of variational parameters Θ in the optimization is illustrated in Fig. 3.

Algorithm 1 ScoreDVI+ for real-world denoising

Input: GMM number K , MC number M , AWGN Gaussian denoisers $\mathcal{G}_{\theta}^{1,\dots,K}(\cdot)$, iterations T , noisy image $\mathbf{y}$
Output: denoised image and its variance

- 1: $t = 1$, initialize Θ
- 2: **while** $t \leq T$ **do**
- 3: **for** $k \leq K$ **do**
- 4: Compute the derivatives of $\mathbb{E}_{q(x_{ik})} \log p(x_{ik})$ in $\mathcal{L}_2$ to $\mu_k(t)$ and $\sigma_k(t)$ based on Eq. (25)
- 5: Compute $\mathcal{L}_1, \mathcal{L}_3, \mathcal{L}_4, \mathcal{L}_5$ and the first item in $\mathcal{L}_2$
- 6: Update Θ using gradient-based methods
- 7: **end for**
- 8: **end while**
- 9: **return** $\bar{\mu}(T)$ and $\bar{\sigma}^2(T)$ based on Eq. (26)

Once the optimization of Eq. (18) is done, we can obtain the mean and variance of $q(X)$ from $q(X|Z)$ and $q(Z)$ as follows:

$$\bar{\mu} = \sum_{k=1}^K \pi_k \odot \mu_k, \bar{\sigma}^2 = \sum_{k=1}^K \pi_k^2 \odot \sigma_k^2 \quad (26)$$

which appear to be the fusion of different variational image posteriors $\mathbf{x}_k$. $\bar{\mu}$ is then utilized as the final denoised image and $\bar{\sigma}^2$ represents the uncertainty of $\bar{\mu}$.

We can apply extra regularizations to variational parameters Θ during the optimization in Alg. 1. Inspired by DIP, we utilize learnable CNNs to output Θ , providing better initialization and regularization of Θ . In each iteration, Θ is obtained from CNNs, which are optimized based on the autograd provided by modern deep-learning frameworks, as the derivatives of Θ can back-propagate to CNNs.

4.4 Noise-aware Assignment of Prior

Observe that the image prior and likelihood are given equal weight in Eq. (18). In practice, we expect the weighting between these two terms to depend on the specific input $\mathbf{y}$. For images with less noise, we would like the reconstruction results to be more biased toward the actual observations. On the other hand, for noisier images, we expect the restorations to be more dependent on the prior information. Therefore, we propose to adaptively adjust the weighting between the image priors and likelihoods based on the level of noise present in the input image. Specifically, we scale $\mathcal{L}_2$ to

$$\mathcal{L}_2 = \lambda(\mathbf{y}) \mathbb{E}_{q(Z)} (\text{KL}(q(X|Z) \| p(X|Z))) \quad (27)$$

where $\lambda(\mathbf{y})$ is a unary weight function depending on $\mathbf{y}$.

To roughly estimate the noise level of $\mathbf{y}$, we use the noise estimation method presented in [77] for generalized signal-dependent noise. Since real-world noise is often spatially correlated, we first split $\mathbf{y}$ into down-scaled

sub-images using a PD operation with a stride of 4. We then estimate the noise variance of local patches in these sub-images and retain the patches with weak texture. The average noise standard deviation δ is then calculated from the remaining patches, providing a rough indication of the noise level presented in $\mathbf{y}$. Based on δ , we design a simple $\lambda(\mathbf{y})$:

$$\lambda(\mathbf{y}) = \begin{cases} \frac{1}{\gamma} & \text{if } \delta < 10 \\ 1 & \text{else if } 10 \leq \delta < 25 \\ \gamma & \text{else if } \delta \geq 25 \end{cases} \quad (28)$$

where $\gamma > 1$ is the global weight of priors.

5 Extension to Other Image Restorations

In Section 4, we used score-based VI to handle real-world denoising tasks. This section shows that the proposed approach can be extended to other image restorations, including linear tasks and non-linear tasks.

5.1 Linear Image Restoration

Given the following linear forward process:

$$\mathbf{y} = A\mathbf{x} + \mathbf{n} \quad (29)$$

where A is the system matrix, and $\mathbf{n}$ is assumed as AWGN with variance σ_w^2 . Similar to denoising, the restoration task here is finding posterior $p(\mathbf{x}|\mathbf{y})$ given degraded $\mathbf{y}$, and we again utilize the variational Gaussian posterior $q(\mathbf{x}) \sim \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2))$ to approximate it. Based on Eq. (1), the negative ELBO for image restoration becomes to:

$$\begin{aligned} -\text{ELBO}_{\text{IR}} &= -\mathbb{E}_{q(\mathbf{x})} \log p(\mathbf{y}) + \text{KL}(q||p) \\ &= -\mathbb{E}_{q(\mathbf{x})} \log -\frac{1}{2\pi\sigma_w^2} \exp\left(-\frac{\|A\mathbf{x} - \mathbf{y}\|^2}{2\sigma_w^2}\right) + \text{KL}(q||p) \\ &= \mathbb{E}_{q(\mathbf{x})} \frac{(A\mathbf{x} - \mathbf{y})^T (A\mathbf{x} - \mathbf{y})}{2\sigma_w^2} + \text{KL}(q||p) + C_1 \\ &= \frac{\|A\boldsymbol{\mu} - \mathbf{y}\|^2 + \text{tr}(A^T \text{Adiag}(\boldsymbol{\sigma}^2))}{2\sigma_w^2} + \text{KL}(q||p) + C_1 \\ &= \|A\boldsymbol{\mu} - \mathbf{y}\|^2 + \text{tr}(A^T \text{Adiag}(\boldsymbol{\sigma}^2)) + \lambda \text{KL}(q||p) + 2\sigma_w^2 C_1 \end{aligned} \quad (30)$$

where $\text{KL}(q||p) = \text{KL}(q(\mathbf{x})||p(\mathbf{x}))$; $\text{tr}(\cdot)$ denotes the trace operation; in the fourth row we utilize the fact that $\mathbb{E}(\mathbf{x}^T A^T A \mathbf{x}) = \boldsymbol{\mu}^T A^T A \boldsymbol{\mu} + \text{tr}(A^T \text{Adiag}(\boldsymbol{\sigma}^2))$; $C_1 = \frac{1}{2} \log \sigma_w^2$ is a constant; $\lambda = 2\sigma_w^2$, and we generally make it tunable in our experiments.

Since this section only considers AWGN, we do not introduce GMM-based likelihood and the corresponding hyperpriors, meaning that $\Theta = \{\boldsymbol{\mu}, \boldsymbol{\sigma}\}$ are all variational parameters to be optimized. Regarding the optimization of $\text{KL}(q(\mathbf{x})||p(\mathbf{x}))$, it has been well introduced in Section 4.1. Here, we focus on the first two terms in Eq. (30) related to A . Particularly, using the cyclic property of the trace results in:

$$\text{tr}(A^T \text{Adiag}(\boldsymbol{\sigma}^2)) = \sum_{i=1}^N \sigma_i^2 \sum_j A_{ji}^2 \quad (31)$$

where A_{ji} represents the element in the j -row and i -column of A .

Different restoration tasks involve different A , resulting in different $\|A\boldsymbol{\mu} - \mathbf{y}\|^2$ and $\text{tr}(A^T \text{Adiag}(\boldsymbol{\sigma}^2))$. This paper mainly considers image deblurring and inpainting tasks.

Image Deblurring. In deblurring, A represents the blur operation and is generally constructed from a blur kernel $\mathcal{K} \in \mathbb{R}^{s \times s}$. Each row of A is composed of all elements in $\mathcal{K}$ and additional 0 values. It is easy to verify that $\sum_j A_{ji}^2$ equals $\|\mathcal{K}\|^2$ when applying convolution with circular boundary conditions. As a result, the general loss in Eq. (30) can be simplified into:

$$\mathcal{L}_{\text{deblur}} = \|\mathcal{K} * \boldsymbol{\mu} - \mathbf{y}\|^2 + \|\mathcal{K}\|^2 \|\boldsymbol{\sigma}\|^2 + \lambda \text{KL}(q||p) \quad (32)$$

where $*$ denotes convolution.

The first two terms in Eq. (32) are differentiable w.r.t. $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$, respectively. As a result, Θ can be optimized via gradient-based algorithms. Meanwhile, it is feasible to ensemble multiple priors as done in Section 4.3 by incorporating constant mixing coefficient $\pi_k = 1/K$ and different priors $p(\mathbf{x}_k)$ into $\mathcal{L}_{\text{deblur}}$, i.e.,

$$\begin{aligned} \mathcal{L}_{\text{deblur}2} &= \sum_{k=1}^K \pi_k (\|\mathcal{K} * \boldsymbol{\mu}_k - \mathbf{y}\|^2 + \|\mathcal{K}\|^2 \|\boldsymbol{\sigma}_k\|^2 \\ &\quad + \lambda \text{KL}(q(\mathbf{x}_k)||p(\mathbf{x}_k))) \end{aligned} \quad (33)$$

where $q(\mathbf{x}_k) = \mathcal{N}(\boldsymbol{\mu}_k, \text{diag}(\boldsymbol{\sigma}_k^2))$, and the final restoration result is given by Eq. (26).

Image Inpainting In inpainting tasks, A represents the mask operation and is constructed from the binary mask $\mathcal{M} \in \mathbb{R}^{N \times 1}$ with 0 values indicating masked pixels. In such a case, $A = \text{diag}(\mathcal{M})$, and $\sum_j A_{ji}^2$ is simply 0 when pixel i is masked or 1 otherwise. That is, $\text{tr}(A^T \text{Adiag}(\boldsymbol{\sigma}^2)) = \|\mathcal{M} \odot \boldsymbol{\sigma}\|^2$. We can then reduce Eq. (30) to:

$$\mathcal{L}_{\text{inpaint}} = \|\mathcal{M} \odot \boldsymbol{\mu} - \mathbf{y}\|^2 + \|\mathcal{M} \odot \boldsymbol{\sigma}\|^2 + \lambda \text{KL}(q||p) \quad (34)$$

5.2 Non-linear Image Restoration

Considering the non-linear degradation process:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}) + \mathbf{n} \quad (35)$$

where $\mathcal{F}(\cdot)$ denotes a non-linear operator, and $\mathbf{n}$ represents AWGN with variance σ_w^2 . Similar to the derivation in Eq. (30), the corresponding negative ELBO for addressing such non-linear problems can be formulated as:

$$\begin{aligned} -\text{ELBO}_{\text{NIR}} &= -\mathbb{E}_{q(\mathbf{x})} \log p(\mathbf{y}) + \text{KL}(q||p) \\ &= \mathbb{E}_{q(\mathbf{x})} \frac{\|\mathcal{F}(\mathbf{x}_l) - \mathbf{y}\|^2}{2\sigma_w^2} + \text{KL}(q||p) + C_1 \\ &\approx \frac{1}{L} \sum_{l=1}^L \|\mathcal{F}(\mathbf{x}_l) - \mathbf{y}\|^2 + \lambda \text{KL}(q||p) + 2\sigma_w^2 C_1 \end{aligned} \quad (36)$$

with $\mathbf{x}_l \sim q(\mathbf{x})$

where Monte Carlo integration with $L = 10$ samples is employed to approximate $\mathbb{E}_{q(\mathbf{x})} \|\mathcal{F}(\mathbf{x}_l) - \mathbf{y}\|^2$, since the non-linear operator $\mathcal{F}(\cdot)$ does not allow for an analytical expansion as in the linear case; the remaining terms are identical to those in Eq. (30).

In this paper, we investigate two representative and practical non-linear restoration tasks, namely High Dynamic Range (HDR) imaging and JPEG restoration. For HDR, the non-linear operator $\mathcal{F}(\cdot)$ is defined as:

$$\mathcal{F}(\mathbf{x}) = \text{clip}(s \times (\mathbf{x} \times 2 - 1), -1, 1)/2 + 0.5 \quad (37)$$

where $\mathbf{x}, \mathcal{F}(\mathbf{x}) \in [0, 1]$, $\text{clip}(\cdot, -1, 1)$ denotes a non-linear intensity truncation function, and s controls the truncation strength.

For JPEG restoration, $\mathcal{F}(\cdot)$ consists of color space conversion, DCT, quantization, and entropy coding, where the quantization operation, parameterized by a quality factor f , introduces strong non-linearity.

6 Experiments

6.1 Experimental Settings

Dataset: Regarding *real-world image denoising*, we evaluate our method on three modalities: Photography Images (PI), Fluorescence Microscopy Images (FMI), and CT images. For PI, we consider three widely-used datasets, namely SIDD [78], PolyU [79], and CC [80]. The SIDD dataset contains natural sRGB images under varying lighting conditions from smartphones, and we evaluate our method using the SIDD validation and benchmark datasets, each of which consists of 1280 patches with size $3 \times 256 \times 256$. The PolyU and CC consist of 100 and 15 natural images, respectively, taken

from diverse commercial camera brands. Each image in these datasets has a size of $3 \times 512 \times 512$.

For FMI, we choose the public FMDD [2], which contains real-world fluorescence images, and we evaluate our method on its mixed test set with raw images, which have the image size of 512×512 . For CT images, we consider the Mayo 2020 dataset, released by Mayo Clinic in 2020 as the ‘‘Low Dose CT Image and Projection Data’’ dataset [81], which includes multiple Low-Dose (LD) abdomen CT images (1mm thickness, 25% dose level) and chest CT images (1.5mm thickness, 10% dose level). The lower the level of radiation doses, the more severe the noise in LDCT images. For evaluation, we selected patients L506 with 526 abdominal images and C295 with 309 chest images from chest scans, respectively. These CT images have sizes 512×512 .

Regarding *image deblurring and inpainting*, we employ the commonly used Set5/Set14 datasets, synthesizing blurry and painted images using Gaussian (isotropic/anisotropic)/motion blur kernels and random texts, respectively. Similarly, *HDR and JPEG restoration* are also evaluated on the Set14 dataset.

Implementation of MMSE Gaussian denoisers:

We obtain blind MMSE Gaussian denoisers using the bias-free (BF) [82] version of 4-level UNet [83]. For natural image denoising, deblurring, inpainting, HDR, and JPEG restoration, we train multiple blind BF-UNet over different clean datasets, including CBSD500 [84], DIV2K [85], and ImageNet [86] validation dataset [87], to learn diverse priors implicitly. For each high-quality dataset, we add AWGN with random standard deviations from $\mathcal{U}(0, 100)$ (assume the intensity range is $[0, 255]$) to clean patches to construct noisy/clean pairs. The BF-UNet is then optimized with the MSE objective using these paired data.

For FMI, we train BF-UNet on grayscale images of the above datasets to capture priors. For CT images, we select over 5,000 Normal-Dose (ND), i.e., high-quality CT images from abdomen images in Mayo 2020 dataset, excluding test patients, and train one blind BF-UNet over AWGN with standard deviations from $\mathcal{U}(0, 25)$.

Implementation of variational parameters:

For PI and FMI, we employ four CNNs to model Θ in Eq. (17), providing extra regularization. We use a 4-level Unet with skip connections to output $(\mu_{1,\dots,K}, \sigma_{1,\dots,K})$. Remaining $(\hat{\alpha}_{1,\dots,K}, \hat{\beta}_{1,\dots,K})$, $\hat{\mathbf{d}}_{1,\dots,K}$, and $\omega_{1,\dots,K}$ are modeled by three five-layer full convolution networks as used in VDN [66], respectively. Details of these networks can be found in SM S2.

For CT denoising, image deblurring, inpainting, HDR, and JPEG restoration, we randomly initialize Θ and optimize it from scratch without using CNNs.

Table 2: Main experimental settings of our method in diverse restoration tasks. Particularly, MC number $M = 1$ is sufficient for all modalities except CT; $K = 1$ and $M = 5$ work fine for CT denoising; large learning rates are chosen when no CNN regularization is applied.

Tasks	PI denoising			FMI denoising	CT denoising		Deblur	Inpaint&HDR&JPEG
Datasets	SIDD	CC	PolyU	FMDD	Abodem	Chest	Set5&14	Set14
GMM number K	3			3	1		3	1
MC number M	1			1	5		1	1
$\beta(\alpha = 1)$	0.02	0.01	0.005	0.02	5e-4	5e-3	NA	NA
Learning rate	1e-3 to 5e-4				1e-1 to 5e-2		1e-2 to 1e-3	
Use CNN reg.?	Yes				No		No	

Selection of hyperprior parameters: Regarding *real-world denoising*, α is fixed to 1 for all datasets. We set $\beta = 0.02$ for SIDD and FMDD, which are known to contain highly noisy images. For CC and PolyU, which have medium and low noise levels, respectively, we set $\beta = 0.01$, and $\beta = 0.005$. For CT images, we set $\beta = 1e - 3$ and $\beta = 5e - 3$ for abdomen and chest images, respectively, as CT images have a larger dynamic range. γ in Eq. (28) is set to 2. The ablation and sensitivity analysis of these hyperparameters are given in Section. We set $\mathbf{d}$ as an all-one vector in Eq. (13), to assign the same prior weight to each component. Regarding *image deblurring and inpainting*, λ is chosen based on noise levels and mask ratios. For *HDR and JPEG restoration*, λ is chosen based on the truncation strength s and the quality factor f .

Optimization details: During optimization, all losses in $\mathcal{L}_{\text{denoise}}$ except $\mathcal{L}_2$ can be expressed as scalar functions in Pytorch [89]. As for $\mathcal{L}_2$, we implement its gradient backpropagation by using a customized *autograd* unit inherited from *torch.autograd.Function*. Its backward function computes and returns the gradients based on Eqs. (11) and (24). Incorporating this unit into the total loss $\mathcal{L}$ enables end-to-end optimization of our method in PyTorch. Eqs. (32) and (34) are optimized in similar ways. We use the Adam optimizer and perform 500 iterations for denoising and 2000 for deblurring, inpainting, HDR, and JPEG restoration. Table 2 summarizes the main settings and parameters for different tasks and datasets.

We evaluate the restoration quality using the Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM) metrics. By convention, PSNR and SSIM of CT images are computed based on the CT window of [-1024, 3072] Hounsfield Unit (HU). All experiments are conducted on an NVIDIA GeForce GTX 2080 Ti GPU.

6.2 Evaluation on PI and FMI Denoising

Comparisons with zero-shot methods. We compare our method against several zero-shot methods,

including BM3D [88], DIP [15], Self2Self [16], PD-denoising [19], NN+denoiser [14], APBSN-single [1], ZS-N2N [17], MASH [54] and Quan’s method [55]. For DIP, we choose the same 4-level UNet. Due to the well-known overfitting issue, we choose the optimal denoising results guided by ground truth (GT) images. Hence, DIP only serves as a reference. For APBSN-single, we adapt the APBSN proposed in [1] to directly denoise a single image. The strides of PD in training and testing are 5 and 2, respectively. For NN+denoiser [14], we choose its best version, i.e., NN+BM3D, for zero-shot denoising. We re-evaluate the performance of NN+BM3D on FMDD, PolyU, and CC without reference to GTs based on their source code. Regarding rest methods, we either use their official codes or adopt their validated results if available. We summarize the quantitative comparisons in Table 3. Qualitative comparisons between different zero-shot methods on SIDD, PolyU, and FMDD are shown in Figs. 4, 6, and 5, respectively. More visual comparisons can be found in SM S3.

On average, our ScoreDVI achieves the best quantitative and qualitative performance among other zero-shot methods on SIDD and FMDD, which contain noisier images. PD-denoising can effectively remove noise, but weedy artifacts are frequently observed in restored images, as indicated in Figs. 4 and 5. Due to the large stride of PD used in training, APBSN-single severely destroys the details of the original images and leaves clear color artifacts, see Fig. 6, or severe blur in Fig. 5. NN+denoiser introduces over-smoothing while removing noise. ZS-N2N and Self2Self perform poorly on SIDD and FMDD, and are unable to eliminate structured noise. Subsequent MASH and Quan’s method demonstrate improved performance over ZS-N2N and Self2Self. In total, our method can denoise images and retain more details, yielding better results. Regarding CC and PolyU, Self2Self achieves the best PSNRs at the cost of hundreds of thousands of training iterations. It also brings over-smoothed reconstructions, as shown in Fig. 6. Instead, our ScoreDVI+ is more efficient while preserving fine textures.

Table 3: Quantitative comparisons (PSNR(dB)/SSIM) of our method and other real-world denoising methods, including (i): zero-shot methods and (ii): dataset-based methods on PI and FMI datasets. The best PSNR/SSIM results **under each method type and dataset** are marked in blue.

Type	Method	SIDD-Val [78]	SIDD-Bench [78]	FMDD [2]	PolyU [79]	CC [80]	Avg.
(i)	BM3D [88]	25.65/0.475	25.65/0.685	30.06/0.771	37.40/0.953	35.19/0.858	30.79/0.748
	DIP [15]	32.11/0.740	—	32.90/0.854	37.17/0.912	35.61/0.912	34.45/0.855
	Self2Self [16]	29.46/0.595	29.51/0.651	30.76/0.695	38.33/0.962	37.44/0.948	33.10/0.770
	PD-denoising [19]	33.97/0.820	33.61/0.894	33.01/0.856	37.04/0.940	35.85/0.923	34.70/0.887
	NN+denoiser [14]	—	33.18/0.895	32.21/0.831	37.66/0.956	36.52/0.943	34.89/0.906
	APBSN-single [1]	30.90/0.818	30.71/0.869	28.43/0.804	29.61/0.897	27.72/0.891	29.47/0.856
	ZS-N2N [17]	25.58/0.433	—	31.61/0.767	36.05/0.916	33.58/0.854	31.71/0.743
	ScoreDVI [69]	34.75/0.856	34.60/0.920	33.10/0.865	37.77/0.959	37.09/0.945	35.46/0.909
	MASH [54]	35.06/0.851	34.78/0.900	33.71/0.882	37.62/0.932	36.87/0.935	35.61/0.900
	Quan’s method [55]	35.31/0.868	35.05/0.922	33.95/0.885	37.88/0.959	37.20/0.948	35.88/0.916
(ii)	ScoreDVI+(Ours)	35.26/0.867	36.01/0.926	33.16/0.854	38.14/0.962	37.19/0.949	35.95/0.912
	APBSN [1]	36.80/0.874	36.91/0.931	31.99/0.836	37.03/0.951	34.88/0.925	35.52/0.903
	CVF-SID [11]	34.81/0.944	34.71/0.917	32.73/0.843	35.86/0.937	33.29/0.913	34.28/0.911
	PUCA [13]	36.40/0.863	36.88/0.929	29.26/0.804	36.71/0.945	35.37/0.925	34.92/0.893

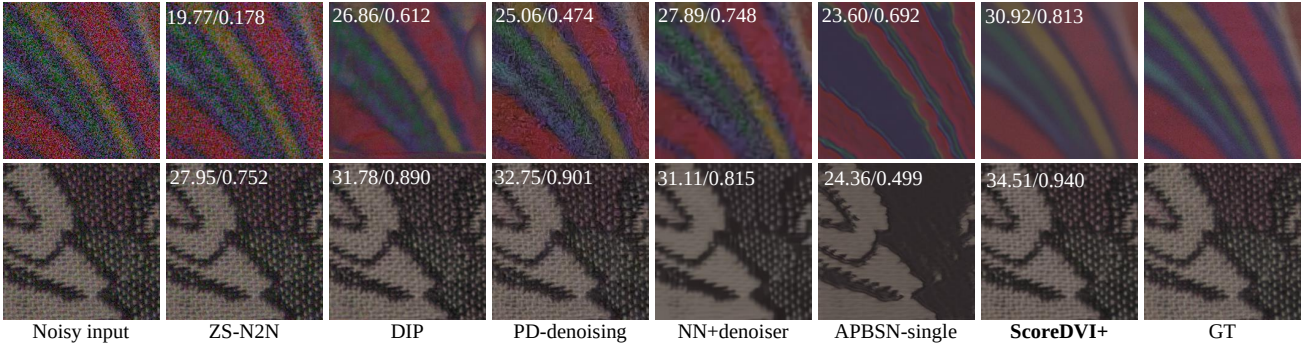


Fig. 4: Visual comparison of our method against other zero-shot methods in SIDD-Val. The PSNR/SSIM results are listed within the images.

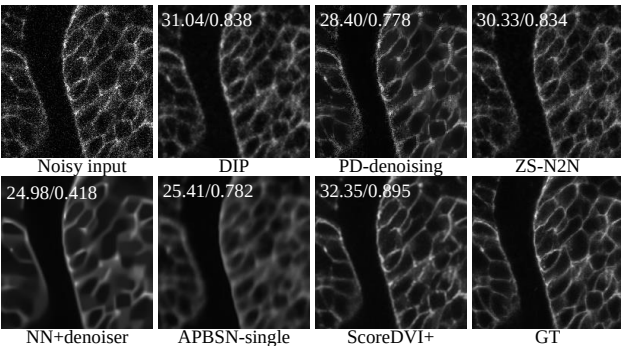


Fig. 5: Visual comparison of zero-shot methods in FMDD. PSNR/SSIM are evaluated on the whole image. The noisy patch is from Confocal_FISH_4.

Comparisons with dataset-based methods. To further verify the superiority of the proposed zero-shot method, we consider three SOTA dataset-based self-supervised denoising methods for comparison, including CVF-SID [11], APBSN [1], and PUCA [13]. For

fair comparisons, these methods are directly trained on respective test sets, without using external data. We present the quantitative results in Table 3, and the visual comparisons can be found in SM S3. The proposed ScoreDVI+ **surpasses** these dataset-based methods *on average*. Although they outperform ours on SIDD-val and -bench, their performance degrades significantly when fewer noisy images are available during training, i.e., PolyU, CC, and FMDD datasets. In contrast, our method requires only a single noisy image and performs consistently over all datasets, making it more practical for real-world applications.

Comparisons with diffusion-based methods. We compare our method with several zero-shot diffusion-based restoration approaches, including GDP [90], DR2 [91], DDRM [27], DPS [29], DiffusionVI [76], RED-diff [41], BIR-D [37], and BIRD [37]. A GAN-based method, i.e., DGP [92], is also included for comparison. The corresponding quantitative results are summarized in Table 4. As shown in Table 4, most diffusion-

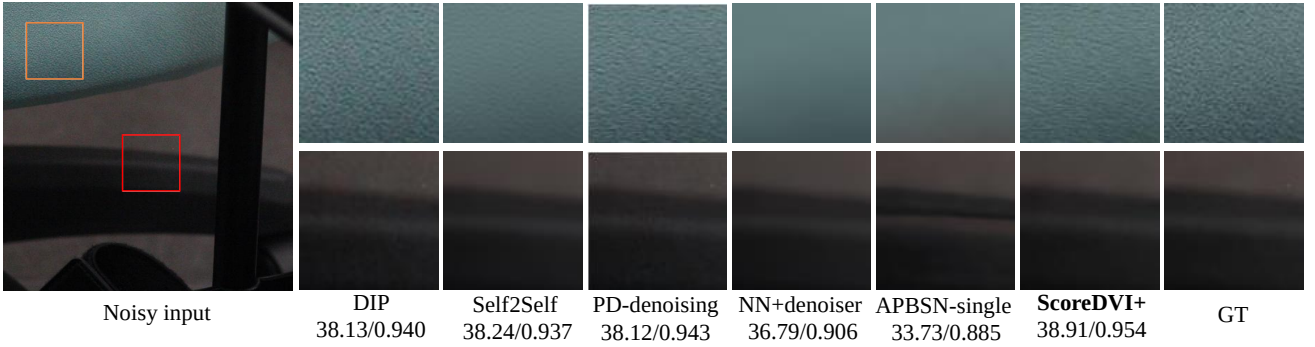


Fig. 6: Visual comparison of our method against other zero-shot methods in PolyU dataset.

Table 4: Quantitative comparisons of our method and diffusion-based methods on real-world image denoising.

Methods	SIDD-Val	FMDD
GDP [90]	27.65/0.615	27.68/0.698
DR2 [91]	32.02/0.728	30.52/0.813
DDRM [27]	33.14/0.796	32.54/0.837
DPS [29]	34.46/0.881	32.26/0.841
DiffusionVI [76]	34.76/ 0.887	33.14/ 0.860
RED-diff [41]	30.00/0.610	29.70/0.665
BIR-D [37]	28.01/0.622	27.80/0.701
BIRD [38]	33.15/0.838	29.68/0.743
DiffPIR [30]	33.63/0.797	32.80/0.855
DGP [92]	23.62/0.306	26.17/0.461
ScoreDVI+	35.26 /0.867	33.15 /0.854

based methods exhibit suboptimal performance in real-world denoising, primarily because they typically assume AWGN rather than complex, realistic noise distributions. In particular, recent methods such as BIR-D and BIRD focus on blind degradation and restoration, but they mainly handle unknown degradation operators (e.g., blur or downsampling) instead of noise. Consequently, their performance on real-world denoising tasks remains limited. Currently, DiffusionVI represents the state-of-the-art diffusion-based approach for real-world noise removal. Our method achieves comparable performance to DiffusionVI, while eliminating the need for explicit score networks and pre-defined posterior sampling procedures, thereby offering greater flexibility. Furthermore, compared with DiffusionVI, our approach converges faster and is more computationally efficient, as demonstrated in Section 6.7. Additional qualitative comparisons are provided in SM S3, Figs. S7–S8.

6.3 Evaluation on CT Image Denoising

We choose several representative CT denoising methods for comprehensive comparison, including unsupervised methods: LIRNet [9] and DML-CT [93]; self-supervised methods: APBSN [1] and Noise2Sim [94]; zero-shot

Table 5: Quantitative comparisons on low-dose CT images. (i): unsupervised/self-supervised methods; (ii): zero-shot methods.

Type	Method	Abdomen	Chest
	LDCT	42.35/0.948	27.47/0.508
(i)	LIRNet [9]	41.65/0.949	35.01/0.822
	DML-CT [93]	44.25/0.965	34.60/0.820
	APBSN [1]	38.86/0.853	31.82/0.773
	Noise2Sim [94]	45.98/0.978	34.47/0.803
(ii)	BM3D [88]	42.66/0.952	29.31/0.642
	DIP [15]	43.54/0.961	33.64/0.771
	ScoreSDE-Med [95]	45.36/0.972	33.40/0.786
	ScoreDVI+	45.95/0.978	35.20/0.820

methods: BM3D [88], DIP [15], and ScoreSDE-Med [95]. For ScoreSDE-Med, we utilize their source code and default parameters to train score networks on ND abdomen and chest images, respectively, from Mayo 2020 dataset. Subsequently, we employ the optimized score network combined with their conditioning and sampling method to generate 10 denoised samples for each LDCT image, and we evaluate their performance based on the average denoised output. We retrain the remaining deep learning-based methods based on their official source codes and our datasets.

We present the quantitative evaluation of different methods on abdomen and chest images in Table 5. Additionally, we provide visual comparisons in Fig. 7 and SM S3. As shown in Table 5, our method outperforms most compared methods quantitatively, except Noise2Sim in abdomen images (0.03dB smaller). Note that, Noise2Sim relies on multiple *adjacent similar slices* for self-supervised training, which has limited application compared with zero-shot methods, and our approach also surpasses in chest images. On the other hand, BM3D and DIP struggle to remove LDCT noise. While ScoreSDE-Med is effective for abdomen images, its performance on chest images is poor, potentially due to suboptimal scores learned on limited ND chest images. In contrast, our method does not rely on any ND

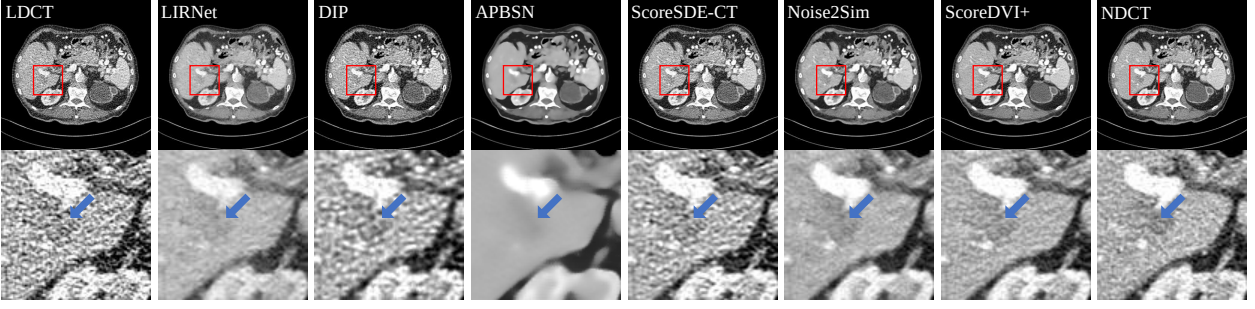


Fig. 7: Visual comparisons of different methods on a low-dose abdomen image. All images are displayed in the CT window of $[-160, 240]$ HU. The blue arrow indicates a lesion area in the NDCT image, which is unclear in the LDCT image. Our method successfully restores this area.

Table 6: Quantitative results of image deblurring (Gaussian blur). (i)-iso: isotropic Gaussian kernel with $k5s10$; (ii)-aniso: anisotropic Gaussian kernel with $k9s[20, 1]$.

Type	Method	Set5	Set14	Type	Method	Set5	Set14
(i)-iso $\sigma_w = 5.1$	DDRM	30.71/0.859	28.72/0.789	(i)-iso $\sigma_w = 12.75$	DDRM	29.01/ 0.824	26.90/ 0.724
	DiffPIR	30.40/0.847	28.18/0.768		DiffPIR	28.31/0.800	26.20/0.695
	BIR-D	22.89/0.576	22.17/0.566		BIR-D	22.83/0.573	22.03/0.550
	BIRD	24.05/0.629	23.13/0.589		BIRD	24.05/0.629	23.09/0.584
	RED-diff	28.19/0.799	26.31/0.706		RED-diff	26.61/0.678	25.07/0.607
	ScoreDVI+	31.07/0.860	28.58/0.783		ScoreDVI+	29.11/0.818	27.06/0.722
(ii)-aniso $\sigma_w = 5.1$	DDRM	29.64/0.837	27.90/0.766	(ii)-aniso $\sigma_w = 12.75$	DDRM	27.80/0.795	25.98/0.697
	DiffPIR	29.53/0.825	27.61/0.754		DiffPIR	27.43/0.774	25.56/0.676
	BIR-D	22.80/0.623	22.12/0.548		BIR-D	22.77/0.612	21.86/0.536
	BIRD	23.19/0.593	22.74/0.569		BIRD	23.15/0.590	22.50/0.560
	RED-diff	26.23/0.749	25.17/0.673		RED-diff	25.49/0.665	24.43/0.601
	ScoreDVI+	30.20/0.845	27.96/0.774		ScoreDVI+	28.38/0.801	26.42/0.715

chest scans during training $\mathcal{G}_\theta(\cdot)$ and is directly applicable to chest images. In addition, denoised CT images from the proposed method exhibit better visual quality, enabling clinical diagnosis as shown in Fig. 7.

6.4 Evaluation on Linear Restoration Tasks

1) **Image deblurring results under isotropic and anisotropic Gaussian blurs.** Following the setting of DDRM, we consider one isotropic Gaussian kernel, denoted as $k5s10$ (i.e., kernel size = 5, kernel std = 10), and one anisotropic Gaussian kernel, denoted as $k9s[20, 1]$ (i.e., kernel size = 9, with kernel std of 20 and 1 along the x - and y -axes, respectively). AWGN with different noise std σ_w is further added to construct blurry images. Similar to the previous subsection, we compare representative diffusion-based approaches, including DDRM, DiffPIR, BIR-D, BIRD, and RED-diff. Quantitative and qualitative deblurring results are reported in Table 6 and Figure 8, respectively.

As shown in Table 6, our method achieves the highest PSNR and SSIM scores across nearly all blur types and noise levels, demonstrating the effectiveness

of ScoreDVI+. It is worth noting that both BIR-D and BIRD rely on projecting latent variables of diffusion models onto the pixel space to incorporate guidance from the observed degraded images, similar to GDP and DPS. Although these methods can handle complex or unknown degradations, they depend on approximate posterior sampling and thus underperform DDRM in linear restoration tasks—an observation also confirmed in the experiments of DPS and GDP. Visual comparisons in Fig. 8 further verify that our method can effectively remove blur and recover sharp image details, whereas DDRM occasionally introduces artifacts. Extra visual results are provided in SM S3, Fig. S9.

2) **Image deblurring results under motion blurs.** Following the setting of DPS, we randomly generate two motion blur kernels (visualized at the bottom-left corners of Fig. 9) and conduct motion deblurring experiments accordingly. As shown in Table 7 and Fig. 9, our method effectively handles complex and severe motion blurs, achieving superior performance compared to DPS and RED-diff.

3) **Image inpainting results.** We use random binary texts as masks to paint clean images. As shown in Fig. 10, our method is effective in inpainting, filling



Fig. 8: Visual results of deblurring under anisotropic Gaussian kernel $k_{9s}[20, 1]$.

Table 7: Quantitative results of image deblurring (Motion blur) on the Set14 dataset. $\sigma_w = 5.1$

Type	DPS	RED-diff	Ours
kernel 1	22.75/0.554	26.32/0.722	25.74/0.699
kernel 2	19.28/0.438	21.98/0.568	23.97/0.639
Avg.	21.02/0.496	24.11/0.645	24.86/0.669

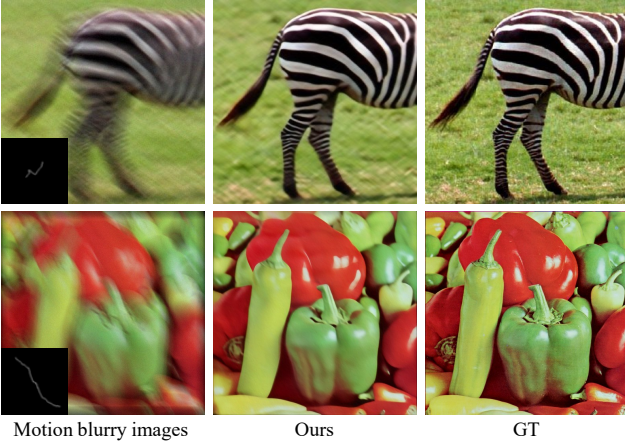


Fig. 9: Visualization of motion deblurring. Motion blur kernels are depicted in the bottom-left corners.

masked areas with reasonable and consistent textures and colors.

6.5 Evaluation on Non-linear Restoration Tasks

Following the setting of RED-diff, the truncation strength s in the forward operator of HDR is set to 2, which synthesizes low dynamic range (LDR) images containing both over-exposed and under-exposed regions, as illustrated in the top-left image of Fig. 11. For JPEG restoration, the quality factor f is set to 10,



Fig. 10: Image inpainting of the proposed ScoreDVI+.

Table 8: Quantitative results of HDR ($s = 2$) and JPEG ($f = 10$) restoration on Set14 dataset

Type	DPS	RED-diff	Ours
HDR	12.24/0.406	24.36/0.835	21.19/0.744
JPEG	8.52/0.171	8.77/0.279	26.19/0.752

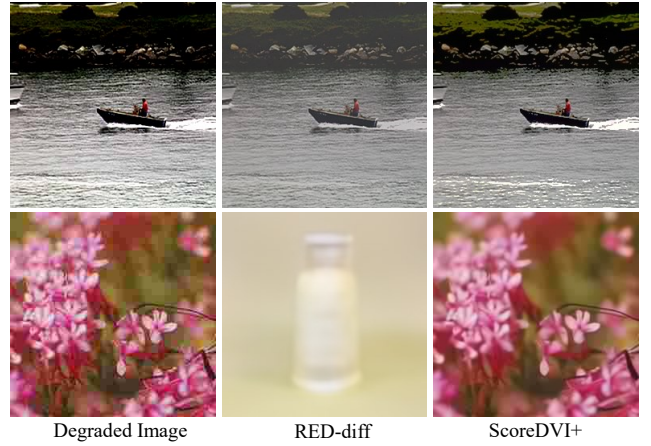


Fig. 11: Visualization of non-linear restorations (Top: HDR, Bottom: JPEG).

leading to noticeable blocking artifacts and color distortions, as shown in the bottom-left of Fig. 11.

As shown in Table 8, DPS exhibits poor performance on these non-linear tasks. RED-diff also fails to handle JPEG restoration, often producing irrelevant content and unrealistic structures. In contrast, the proposed ScoreDVI+ achieves higher PSNR and SSIM values than RED-diff and DPS on average. Fig. 11 further demonstrates that our method can effectively enhance the dynamic range in over-/under-exposed regions (top part) and remove blocking artifacts in JPEG-compressed images (bottom part), highlighting its flexibility and superiority in addressing non-linear tasks.

6.6 Ablation studies

Here, we conduct several ablation studies to analyze the proposed method in depth.

Effectiveness of score priors. We ablate the score prior extracted from Gaussian denoiser $\mathcal{G}_\theta^k(\cdot)$, which is

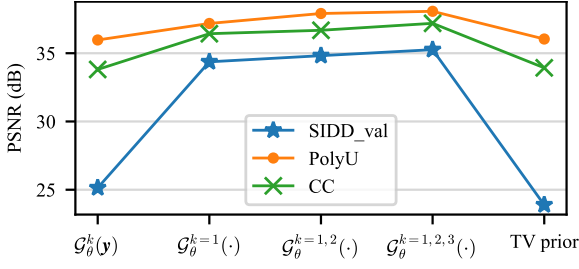


Fig. 12: Effects of using $\mathcal{G}_\theta^k(\mathbf{y})$ (1st column), implicit scores from $\mathcal{G}_\theta^k(\cdot)$ with different K (2nd to 4th columns) and the TV prior (5th column).

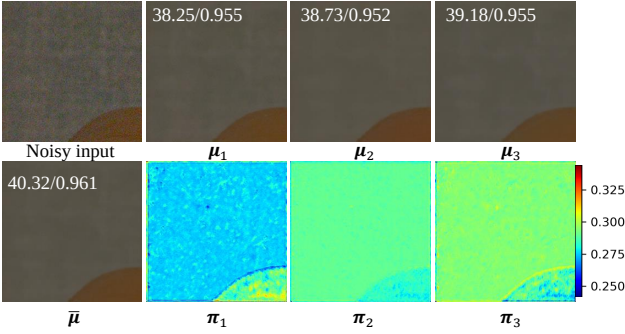


Fig. 13: Visualization of different μ_k and π_k used for posteriors' ensemble.

critical to variational optimization in VI. To first verify that $\mathcal{G}_\theta^k(\cdot)$ indeed serves as (implicit) priors rather than direct denoisers to real-world images, we check the denoising performance of using $\mathcal{G}_\theta^k(\mathbf{y})$ and present results in Fig. 12. It is clear that $\mathcal{G}_\theta^k(\mathbf{y})$ performs poorly in handling real-world complex images, a well-known issue of deep denoisers. When we incorporate scores from $\mathcal{G}_\theta^k(\cdot)$ into the variational optimization, the real-world denoising capacity is dramatically enhanced. See large PSNR gains from the first column to the second column in Fig. 12. In addition, the denoising performance consistently improves as we use more K , i.e., more denoisers $\mathcal{G}_\theta^k(\cdot)$. Fig. 13 indicates that using multiple priors allows an ensemble of different variational image posteriors, achieving better quality than a single prior.

Besides, when $\mathcal{L}_2$ is replaced with a pairwise Markov random field (or TV) prior during the optimization, corresponding PSNRs/SSIMs drop substantially as shown in the last column Fig. 12, further demonstrating the effectiveness of the proposed score priors.

Furthermore, to systematically investigate the impact of Gaussian denoisers $\mathcal{G}_\theta^k(\cdot)$ on real-world denoising performance, we train $\mathcal{G}_\theta^k(\cdot)$ models with varying numbers of parameters. Specifically, we adjust the base channel number (n_{bc}) of the UNet architecture to obtain multiple $\mathcal{G}_\theta^k(\cdot)$ variants, and then conduct real-world denoising experiments on SIDD-Val. The quan-

Table 9: Effects of $\mathcal{G}_\theta^k(\cdot)$ with different base channel numbers (n_{bc}) on real-world denoising performance

n_{bc}	8	16	32	64
Params.(M)	0.012	0.048	1.92	7.7
SIDD-Val (PSNR)	33.34	34.58	35.17	35.26

Table 10: Ablation of different likelihood functions for real-world denoising.

Dataset	(i)	(ii)	Ours
SIDD-Val	24.67/0.370	34.71/0.869	35.26/0.867
PolyU	36.08/0.924	37.88/0.956	38.14/0.962

Table 11: Ablation of score refinement under two kinds of blind denoisers. R denotes refinement

Priors	Type	SIDD_val	Abdomen
BF-DnCNN	w/o R	34.75/0.856	42.19/0.958
	with R	35.04/0.860	45.61/0.975
BF-UNet	w/o R	31.93/0.757	42.95/0.960
	with R	35.26/0.867	45.95/0.978

titative results corresponding to different denoisers are summarized in Table 9, which shows that as the number of network parameters decreases, the real-world denoising performance gradually declines but without a significant drop. Even when $\mathcal{G}_\theta^k(\cdot)$ contains only 0.012M parameters ($n_{bc} = 8$), the resulting PSNR decreases by less than 2 dB compared to the denoiser with 7.7M parameters ($n_{bc} = 64$). This observation indicates that even a lightweight Gaussian denoiser remains effective within the proposed optimization framework.

Effectiveness of non-*i.i.d.* GMM. To demonstrate the effectiveness of the proposed likelihood, we consider two alternative ways: (i): fixed GMM $p(y_i|X, \Phi, \Omega) = \sum_{k=1}^K \omega_{ik} \mathcal{N}(x_{ik}, \frac{\beta_0}{\alpha_0})$ with constant α and β ; (ii): using GMM $p(y_i|X, \Phi, \Omega) = \sum_{k=1}^K \omega_{ik} \mathcal{N}(x_{ik}, \phi^{-1})$ with the scalar precision ϕ inferred during VI. Quantitative comparisons of these three ways to model the likelihood are presented in Table 10. Comparing the results from (i) and (ii) reveals that inferring the posterior precision during VI is critical for denoising as specifying accurate α and β for real-world noisy images is difficult. In addition, as real-world images contain signal-dependent noise, incorporating spatially varying precision into the likelihood further enhances denoising performance.

Effectiveness of score refinement. We consider $\mathcal{G}_\theta^k(\cdot)$ implemented by two different networks: BF-DnCNN used in ScoreDVI [69] and BF-UNet used in this paper. In Table 11, we report denoising results with (i.e., Eq. (12)) and without (i.e., Eq. (7)) the score refinement strategy. Table 11 suggests that score refine-

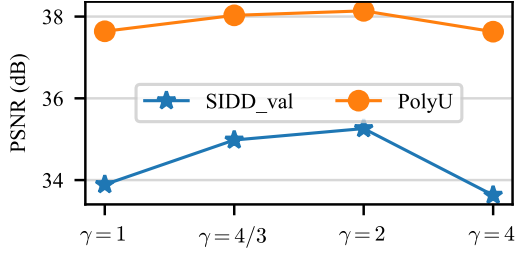


Fig. 14: Effects of prior assignment coefficient γ on real-world denoising.

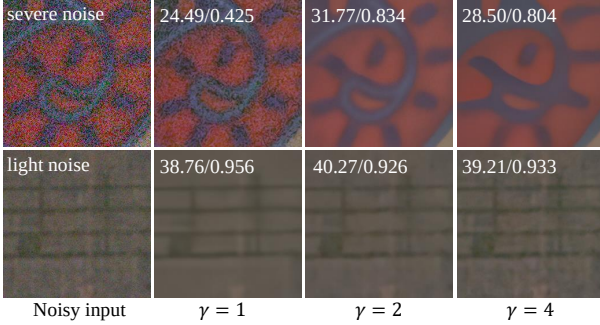


Fig. 15: Visualization of denoised images w.r.t. different γ for severe (top row) and light (bottom row) noises.

ment consistently improves the denoising performance under the two denoisers. Particularly, denoising LDCT images (and SIDD images under BF-UNet) without adopting this strategy shows poor quantitative metrics as implied in the last row of Table 11, indicating the instability issue of optimization with Eq. (7), which is successfully addressed by Eq. (12).

Effectiveness of adaptive prior assignment. Fig. 14 depicts the results of our method with different $\lambda(\mathbf{y})$, also γ . From Figs. 14 and 15, we can observe that: when $\gamma = 1$, it means no adaptive prior assignment, and the corresponding denoised images either contain residual noise for the noisier image (top images of Fig. 15) or are over-smooth for light noise (bottom images of Fig. 15); with the increase of γ , the noise is gradually removed (top images), and details emerge (bottom images); when γ becomes too large, either image priors or actual observations dominate the optimization, and the denoised images show blurriness or become noisy. As a result, we set $\gamma = 2$ in the main experiments.

6.7 Further Analyses

Sensitivity analysis. Our method necessitates two hyperparameters, i.e., α and β , which define the prior variance of noisy images. As shown in Table 2, we empirically assign each test dataset a scalar β to account for

Table 12: Sensitivity analysis of β .

Error of β	SIDD_val	PolyU
-50%	34.06/0.813	37.80/0.955
-25%	35.10/0.855	38.14/0.962
0%	35.26/0.867	38.14/0.962
+25%	35.28/0.871	38.08/0.961
+50%	35.17/ 0.872	38.10/0.962

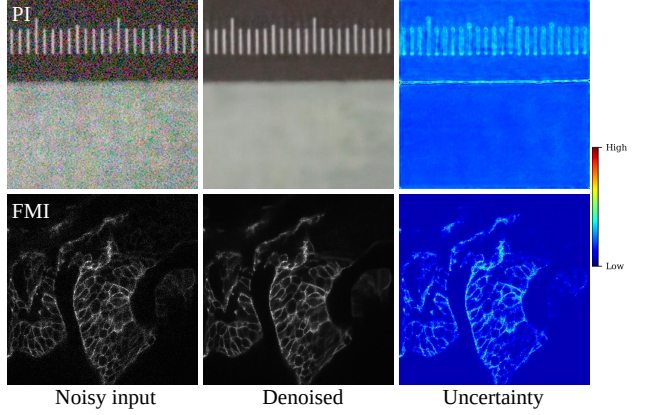


Fig. 16: Visualization of denoised results and uncertainties $\bar{\sigma}^2$ for PI (top row) and FMI (bottom row).

our prior knowledge about the noise level of the whole dataset. Here, we present the sensitivity analysis of β (α is fixed to 1) in Table 12, where different β are chosen to account for the error. As shown in Table 12, the quantitative metrics resulting from different β change slightly, indicating that the proposed method is robust to different parameter selections.

Uncertainty quantification (UQ). Our method enables estimating the variational image posterior variance $\bar{\sigma}^2$, which is identified as the data uncertainty. Pixel-wise $\bar{\sigma}^2$ represents the uncertainty around the variational image posterior mean $\bar{\mu}$ and implies the relative credibility of the denoised image, which is important to safety-critical scenarios. We provide two variance maps of PI and CT images in Fig. 16. As observed in Fig. 16, denoised images with uniform or smooth areas have smaller variances (also uncertainty) as these areas can be reliably restored. Larger variance generally focuses on edges and complex textures that contain more high-frequency components. Because denoising algorithms often struggle to restore these challenging areas, it is reasonable that more uncertainty will be generated.

Additionally, we conduct supplementary experiments to evaluate the reliability of the estimated uncertainty. In particular, we examine the calibration performance of the estimated uncertainty, which is a widely adopted measure in UQ [96]. The detailed procedures

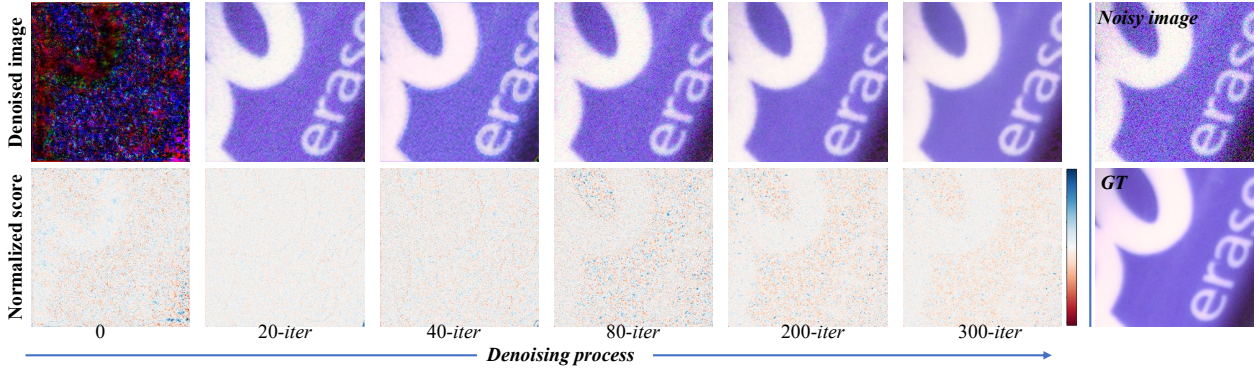
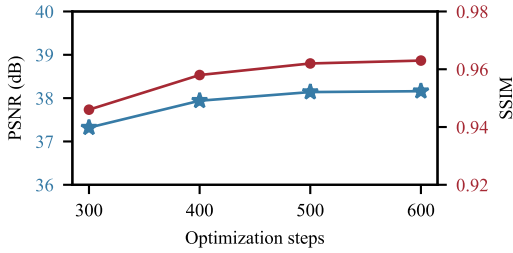
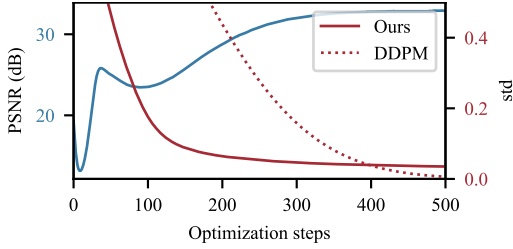


Fig. 17: Visualization of the denoising process on real-world noisy images. Upper: intermediate denoised images; bottom: intermediate scores (i.e., Eq. (12)).



(a) Denoising performance on PolyU w.r.t. different optimization steps T



(b) Convergence of the proposed variational optimization on a single image.

Fig. 18: Empirical convergence of the proposed method

and results are provided in SM S4, which indicate that the estimated uncertainties on SIDD-Val and FMDD are well-calibrated, where the expected confidence levels align with the empirical observations.

Denoising process. Fig. 17 depicts the denoising process for one real-world noisy image (from SIDD-Val), including extracted scores (normalized for visualization) and intermediate denoising means at different iterations, to better understand the proposed method. It can be observed that scores (i.e., optimization gradients from Gaussian denoisers) focus on edges at early optimization stages, and then pay attention to background and noisy areas at later stages, analogous to score-based generative models that first generate semantics and then denoise.

Convergence. We study different iteration numbers T at the dataset and image levels, and the corresponding results are plotted in Fig. 18. As shown in Fig. 18a, increasing T results in the convergence of performance and does not show overfitting issues. And $T = 500$ is sufficient to balance the performance and efficiency *on average* for real-world denoising.

We also analyze the convergence of our method during the optimization. As shown by the solid red plot in Fig. 18b, the average std of the posterior rapidly decreases upon optimization begins, and remains almost steady after 400 iterations. Meanwhile, the PSNR of the denoised image gradually rises and finally stays constant (see the blue plot). As a comparison, we draw the pre-defined stds of DDPM [23] in Fig. 18b (the dotted red line), with a total of 500 diffusion steps for alignment. We can observe that our adaptively updated std of the variational image posterior converges faster than the pre-defined one in DDPM, demonstrating the superiority of the proposed approach.

Complexity. We report the inference time, model parameters, and FLOPs of several deep learning-based zero-shot denoising methods in Table 13, where parameters and FLOPs are computed from the involved deep models in each method. Comparing Table 3 (the denoising performance) and Table 13 reveals that our method well balances the effectiveness and efficiency. Notably, our method is more efficient than the SOTA DiffusionVI, indicating the superiority of the proposed score-driven variational optimization.

7 Conclusion

In this paper, we propose ScoreDVI+, a novel variational inference method that utilizes implicit score priors from MMSE Gaussian denoisers for variational optimization, and incorporates the Non-*i.i.d* Gaussian mix-

Table 13: Efficiency comparisons of different methods under the input size $256 \times 256 \times 3$. Test on Nvidia 2080Ti.

Method	Time(s)	Param.(M)	FLOPs(G)
DIP	146.2	13.4	31.06
Self2Self	3546.5	1.0	9.55
PD-denoising	0.36	0.7	46.94
NN+denoiser	897.6	13.4	31.06
APBSN-single	121.4	3.66	234.63
DDRM	5	552.81	1114.21
DiffusionVI	230	552.81	1114.21
ScoreDVI+	22	21.20	79.23

ture likelihood model for real-world image denoising. A score refinement strategy and noise-aware priors assignment scheme are additionally developed to improve the optimization stability and denoising performance, respectively. The score-based VI is further extended to linear and non-linear restoration tasks, including image deblurring, inpainting, HDR, and JPEG restoration. The proposed method generally shows better performance than other zero-shot and dataset-based restoration methods on various tasks and benchmark datasets.

The proposed method has two limitations. First, as shown in Table 13, ScoreDVI+ costs about 20 seconds to process a single noisy image, which may not be suitable for real-time tasks, such as autonomous driving. Second, when applied to real-world degradations beyond noises, ScoreDVI+ requires accurate system matrices or forward operators, which may not be available in practice. Future research could focus on improving computational efficiency and extending the framework toward blind or semi-blind restoration, where the forward model is partially known or learned jointly with the image.

8 Data Availability

All data generated or analysed during this study are included in published articles (and their additional files). The SIDD dataset [78] has been deposited in the following link: <https://abdokamel.github.io/sidd/>. The FMDD dataset [2] has been deposited in the following link: <https://github.com/yinhaoz/denoising-fluorescence>. The PolyU dataset [79] has been deposited in the following link: <https://github.com/csjunxu/PolyU-Real-World-Noisy-Images-Dataset>. The CC dataset [80] has been deposited in the following link: <http://snam.ml/research/ccnoise>. The LDCT dataset [81] has been deposited in the following link: <https://www.cancerimagingarchive.net/collection/ldct-and-projection-data/>. The Set5/Set14 datasets

have been deposited in the following link: https://figshare.com/articles/dataset/BSD100_Set5_Set14_Urban100/21586188

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